1. Overview of Action Matching

Action Matching: Learning Stochastic Dynamics from Samples is a method for learning a rich family of dynamics using only independent samples from its time evolution.

We will emphasize Entropic Action Matching. Consider that the dynamics of the system evolve according to the following stochastic differential equation (SDE):

$$\mathrm{d}X_t = v_t(x)\mathrm{d}t + \sigma_t \mathrm{d}W_t, \qquad X_0 = x$$

where $v_t : [0,1] \times \mathbb{R}^d \to \mathbb{R}^d$ is the continuous drift of the process, $\sigma_t : [0,1] \to \mathbb{R}$ is the volatility, and $(W_t)_{t\in[0,1]}$ is the standard Brownian motion process on $\mathbb{R}^d$.

The density of the system evolves according to the Fokker-Planck equation:

$$\partial_t q_t = -\nabla \cdot (q_t v_t) + \frac{\sigma_t^2}{2}\Delta q_t$$

Consider a continuous dynamic with the density evolution of q_t, and suppose σ_t is given. Then there exists a unique (up to a constant) function $s_t^*(x)$ called entropic action such that the vector field $v_t^*(x) = \nabla s_t^*(x)$ and q_t satisfy he Fokker-Planck equation:

$$\partial_t q_t = -\nabla \cdot (q_t \nabla s_t^*) + \frac{\sigma_t^2}{2}\Delta q_t$$

Using this result the problem of learning the dynamics can be boiled down to learning the unique vector field $\nabla s_t^*(x)$, using only samples from q_t. Thus we restrict the search space to the family of gradient vector fields:

$$S_t = \{\nabla s_t \mid s_t : \mathbb{R}^d \to \mathbb{R}\}, \quad t \in [0,1]$$

Let $s_t(x)$ be parameterized by a neural network which approximates the optimal $s_t^*(x)$ using the following entropic action gap objective:

$$\mathrm{eAG}(s, s^*) = \frac{1}{2}\int_0^1 \mathbb{E}_{q_t(x)}\left[\|\nabla s_t(x) - \nabla s_t^*(x)\|^2\right] \mathrm{d}t$$

For an arbitrary variational action s, the entropic action gap can be decomposed as the sum of an intractable constant K_{eAM} and a tractable term $L_{eAM}(s)$:

$$\mathrm{eAG}(s, s^*) = K_{eAM} + L_{eAM}(s)$$

where the tractable loss $L_{eAM}(s)$ is the entropic Action Matching objective, which we need to minimize:

$$L_{eAM}(s) = \mathbb{E}_{q_0(x)}[s_0(x)] - \mathbb{E}_{q_1(x)}[s_1(x)] + \int_0^1 \mathbb{E}_{q_t(x)}\left[\frac{1}{2}\|\nabla s_t(x)\|^2 + \partial_t s_t(x) + \frac{\sigma_t^2}{2}\Delta s_t\right] \mathrm{d}t$$

Algorithm 1: Entropic Action Matching

Input: data samples $\{X_t^j\}_{j=1}^{N_t}$, with $x_t^j \sim q_t(x)$, volatility σ_t, and parametric model $s_t(x;\theta)$

while *Not Converged* **do**

sample endpoints $\{X_0^i\}_{i=1}^n \sim q_0(x)$, $\{X_1^i\}_{i=1}^n \sim q_1(x)$

sample time $\{t^i\}_{i=1}^n \sim U[0,1]$

get intermediate samples $\{X_{t^i}^i\}_{i=1}^n \sim q_t(x)$

$L_{01} = \frac{1}{n}\sum_{i=1}^n [s_0(X_0^i;\theta) - s_1(X_1^i;\theta)]$

$$L_{eAM}(\theta) = L_{01} + \frac{1}{n}\sum_{i=1}^n \left[\frac{1}{2}\|\nabla s_{t^i}(X_{t^i}^i;\theta)\|^2 + \partial_t s_{t^i}(X_{t^i}^i;\theta) + \frac{\sigma_{t^i}^2}{2}\Delta s_{t^i}(X_{t^i}^i;\theta)\right]$$

Update θ using $\partial_\theta L_{eAM}$

Output: Trained entropic action $s_t(x;\theta^*)$

Now, we are going to detail the connection of Entropic Action Matching with the EOT problem. The EOT objective can be written as follows:

$$\frac{1}{2}W_e(p_0,p_1) = \inf_{(q_t,v_t)}\left\{\int_0^1 \frac{1}{2}\,\mathbb{E}_{q_t(x)}\left[\|v_t(x)\|^2\right]\mathrm{d}t\right\}$$

under the Fokker-Planck equation constraint $\partial_t q_t = -\nabla\cdot(q_t v_t) + \sigma_t^2/2\;\Delta q_t$, and mariginal constraints $(q_0,q_1) = (p_0,p_1)$.

Since we fix the density path q_t via the provided samples, we omit the optimization over q_t and the marginal constraints. The problem reduces to finding the optimal vector field that minimizes the entropic kinetic energy $\mathcal{K}_{eAM}$:

$$\mathcal{K}_{eAM} = \inf_{v_t}\int_0^1 \frac{1}{2}\,\mathbb{E}_{q_t(x)}\left[\|v_t(x)\|^2\right]\mathrm{d}t$$

subject to the Fokker-Planck constraint.

To solve this, we introduce a Lagrange multiplier s_t to enforce the Fokker-Planck equation, yielding the min-max Lagrangian:

$$\mathcal{K}_{eAM} = \sup_{s_t}\inf_{v_t}\left\{\int_0^1 \frac{1}{2}\,\mathbb{E}_{q_t(x)}\left[\|v_t(x)\|^2\right]\mathrm{d}t + \int_0^1\int_{\mathbb{R}^d} s_t(x)\left(\partial_t q_t + \nabla\cdot(q_t v_t) - \frac{\sigma_t^2}{2}\Delta q_t\right)\mathrm{d}x\mathrm{d}t\right\}$$

By integrating by parts we derive:

$$\mathcal{K}_{eAM} = \sup_{s_t}\inf_{v_t}\left\{\int_{\mathbb{R}^d} s_1 q_1 \mathrm{d}x - \int_{\mathbb{R}^d} s_0 q_0 \mathrm{d}x + \int_0^1\int_{\mathbb{R}^d} q_t\left[\frac{1}{2}\|v_t\|^2 - \langle\nabla s_t, v_t\rangle - \partial_t s_t - \frac{\sigma_t^2}{2}\Delta s_t\right]\mathrm{d}x\mathrm{d}t\right\}$$

The only term that involves v_t is:

$$\begin{aligned} L_v &= \frac{1}{2}\|v_t\|^2 - \langle \nabla s_t, v_t \rangle \\ &= \frac{1}{2}(\|v_t\|^2 - 2\langle \nabla s_t, v_t \rangle + \|\nabla s_t\|^2) - \frac{1}{2}\|\nabla s_t\|^2 \\ &= \frac{1}{2}\|v_t - \nabla s_t\|^2 - \frac{1}{2}\|\nabla s_t\|^2 \end{aligned}$$

Thus for a given action s_t L_v minimized for $v_t = \nabla s_t$, thus we can skip the optimization over v_t and derive:

$$\mathcal{K}_{eAM} = \sup_{s_t} \left\{ \int_{\mathbb{R}^d} s_1 q_1 \mathrm{d}x - \int_{\mathbb{R}^d} s_0 q_0 \mathrm{d}x - \int_0^1 \int_{\mathbb{R}^d} q_t \left[\frac{1}{2} \|\nabla s_t\|^2 + \partial_t s_t + \frac{\sigma_t^2}{2} \Delta s_t \right] \mathrm{d}x \mathrm{d}t \right\}$$

By minimizing the negative objective we derive the entropic action matching loss:

$$L_{eAM} = \inf_{s_t} \left\{ \int_{\mathbb{R}^d} s_0 q_0 \mathrm{d}x - \int_{\mathbb{R}^d} s_1 q_1 \mathrm{d}x + \int_0^1 \int_{\mathbb{R}^d} q_t \left[\frac{1}{2} \|\nabla s_t\|^2 + \partial_t s_t + \frac{\sigma_t^2}{2} \Delta s_t \right] \mathrm{d}x \mathrm{d}t \right\}$$

2. Expanding Action Matching to the SB Problem

Deriving the minimization objective

Let $\Omega = \mathcal{C}([0,1], \mathbb{R}^d)$ be the path space and X_t be the canonical coordinate process. Define the reference measure R under which X_t evolves according to the reference drift $f_t(x) : [0,1] \times \mathbb{R}^d \to \mathbb{R}^d$:

$$\mathrm{d}X_t = f_t(X_t)\mathrm{d}t + \sigma_t \mathrm{d}W_t^R$$

where $\sigma_t > 0$ and $(W_t^R)_{t\in[0,1]}$ is the standard Brownian motion on $\mathbb{R}^d$ under the reference R.

Define the controlled measure P under which X_t evolves according to the drift $u_t(x) : [0,1] \times \mathbb{R}^d \to \mathbb{R}^d$:

$$\mathrm{d}X_t = u_t(X_t)\mathrm{d}t + \sigma_t \mathrm{d}W_t^P$$

where $(W_t^P)_{t\in[0,1]}$ is the standard brownian motion under the measure P.

To evaluate the $\mathrm{KL}(P \mid R)$ we need the Radon-Nikodym derivative $\mathrm{d}P/\mathrm{d}R$. First we isolate the Brownian motion under the reference R:

$$\mathrm{d}W_t^R = \frac{1}{\sigma_t}(\mathrm{d}X_t - f_t(X_t)\mathrm{d}t)$$

Substituting the dynamics under the controlled measure P yield:

$$\begin{aligned}\mathrm{d}W_t^R &= \frac{1}{\sigma_t}[u_t(X_t)\mathrm{d}t + \sigma_t \mathrm{d}W_t^P - f_t(X_t)\mathrm{d}t] \\ &= \mathrm{d}W_t^P + \frac{u_t(X_t) - f_t(X_t)}{\sigma_t}\,\mathrm{d}t\end{aligned}$$

Let the scaled drift of u_t defined as v_t meaning:

$$v_t(X_t) = \frac{u_t(X_t) - f_t(X_t)}{\sigma_t}$$

By the Girsanov's Theorem, assuming that Novikov's condition hold:

$$\frac{\mathrm{d}P}{\mathrm{d}R} = \exp\left(\int_0^1 v_t(X_t)\cdot \mathrm{d}W_t^R - \frac{1}{2}\int_0^1 \|v_t(X_t)\|^2 \mathrm{d}t\right)$$

The KL divergence between the controlled path measure P and the reference path measure R is defined as:

$$\mathrm{KL}(P \mid R) = \mathbb{E}_P\left[\log\frac{\mathrm{d}P}{\mathrm{d}R}\right] = \mathbb{E}_P\left[\int_0^1 v_t(X_t)\cdot \mathrm{d}W_t^R - \frac{1}{2}\int_0^1 \|v_t(X_t)\|^2 \mathrm{d}t\right]$$

We substitute the expression of $\mathrm{d}W_t^R$ from above:

$$\mathrm{KL}(P \mid R) = \mathbb{E}_P\left[\int_0^1 v_t(X_t)\cdot \mathrm{d}W_t^P + \frac{1}{2}\int_0^1 \|v_t(X_t)\|^2 \mathrm{d}t\right]$$

But W_t^P is a standard brownian motion under P, thuse the Ito integral $\int_0^1 v_t(x)\mathrm{d}W_t^P$ is a true martingale, so it's expression is zero. This leaves:

$$\mathrm{KL}(P \mid R) = \mathbb{E}_P\left[\int_0^1 \frac{1}{2\sigma_t^2}\|u_t(X_t) - f_t(X_t)|^2 \mathrm{d}t\right]$$

By Tonelli's theorem we swap the expectation and the time integral:

$$\mathrm{KL}(P \mid R) = \int_0^1 \frac{1}{2\sigma_t^2}\mathbb{E}_P[\|u_t(X_t) - f_t(X_t)\|^2]\mathrm{d}t$$

Let $p_t(x)$ be the marginal probability density of X_t under P. Then the expecation can be written as an integral over the state space:

$$\mathrm{KL}(P \mid R) = \int_0^1 \frac{1}{2\sigma_t^2}\,\mathbb{E}_{p_t(x)}[\|u_t(X_t) - f_t(X_t)\|^2]\,\mathrm{d}t$$

Therefore the SB problem can be formulated as:

$$\mathcal{K}_{\text{SB}} = \inf_{u_t, p_t} \left\{ \int_0^1 \frac{1}{2\sigma_t^2} \mathbb{E}_{p_t(x)}[\|u_t(X_t) - f_t(X_t)\|^2] \, \mathrm{d}t \right\}$$

This minimization is not perform over the set (u_t, p_t) independently. Let ϕ be a smooth, compactly supported test function. By the Ito lemma applied to $\phi(X_t)$ under the controlled measure P we get:

$$\mathrm{d}\phi(X_t) = \nabla\phi(X_t) \cdot u_t(X_t)\mathrm{d}t + \nabla\phi(X_t) \cdot \sigma_t \mathrm{d}W_t^P + \frac{\sigma_t^2}{2}\Delta\phi(X_t)\mathrm{d}t$$

Taking the expecation on both sides, the martingale term vanishes:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}_P[\phi(X_t)] = \mathbb{E}_P\left[u_t(X_t) \cdot \nabla\phi(X_t) + \frac{\sigma_t^2}{2}\Delta\phi(X_t) \right]$$

Rewritting the expecation as integral against the marginal density $p_t(x)$:

$$\int_{\mathbb{R}^d} \phi(x)\partial_t p_t(x)\mathrm{d}x = \int_{\mathbb{R}^d} \left(u_t(x) \cdot \nabla\phi(x) + \frac{\sigma_t^2}{2}\Delta\phi(x) \right) p_t(x)\mathrm{d}x$$

Apply integration by parts on the right-hand-side and grouping everything together yields:

$$\int_{\mathbb{R}^d} \phi(x) \left[\partial_t p_t(x) + \nabla \cdot (p_t(x)u_t(x)) - \frac{\sigma_t^2}{2}\Delta p_t(x) \right] \mathrm{d}x = 0$$

Since this hold for all arbitrary test function ϕ defined above we derive the Fokker-Planck equation:

$$\partial_t p_t(x) = -\nabla \cdot (p_t(x)u_t(x)) + \frac{\sigma_t^2}{2}\Delta p_t(x)$$

Thus the intractable path-space SB problem translates to the following:

$$\mathcal{K}_{\text{SB}} = \inf_{u_t} \left\{ \int_0^1 \frac{1}{2\sigma_t^2}\mathbb{E}_{p_t(x)}[\|u_t(X_t) - f_t(X_t)\|^2]\mathrm{d}t \right\}, \quad \text{s.t}$$

$$\partial_t p_t(x) = -\nabla \cdot (p_t(x)u_t(x)) + \frac{\sigma_t^2}{2}\Delta p_t(x)$$

Solving the minimization objective

To solve the constrainted optimization problem we introduce a scalar Langrangian multiplier function $s_t(x)$ to enforce the Fokker-Planck constraint everywhere in state and time:

$$\mathcal{K}_{\text{SB}} = \sup_{s_t} \inf_{u_t} \left\{ \int_0^1 \frac{1}{2\sigma_t^2} \mathbb{E}_{q_t(x)}[\|u_t(x) - f_t(x)\|^2] \mathrm{d}t \right. \\ \left. + \int_0^1 \int_{\mathbb{R}^d} s_t(x) \left(\partial_t p_t + \nabla \cdot (p_t u_t) - \frac{\sigma_t^2}{2} \Delta p_t(x) \right) \mathrm{d}x \mathrm{d}t \right\}$$

By integrating by part each term on the second integral we derive:

$$\int_0^1 \int_{\mathbb{R}^d} s_t \partial_t p_t \mathrm{d}x \mathrm{d}t = \mathbb{E}_{p_1(x)}[s_1] - \mathbb{E}_{p_0(x)}[s_0] - \int_0^1 \mathbb{E}_{p_t(x)}[\partial_t s_t] \mathrm{d}t$$
$$\int_0^1 \int_{\mathbb{R}^d} s_t \nabla \cdot (p_t u_t) \mathrm{d}x \mathrm{d}t = - \int_0^1 \mathbb{E}_{p_t(x)}[\nabla s_t \cdot u_t] \mathrm{d}t$$
$$- \int_0^1 \int_{\mathbb{R}^d} s_t \frac{\sigma_t^2}{2} \Delta p_t \mathrm{d}x \mathrm{d}t = - \int_0^1 \mathbb{E}_{p_t(x)} \left[\frac{\sigma_t^2}{2} \Delta s_t \right]$$

Substituting back those results we get the Langrangian:

$$\mathcal{L}(u_t, s_t) = \mathbb{E}_{p_1(x)}[s_1(x)] - \mathbb{E}_{p_0(x)}[s_0(x)] \\ + \int_0^1 \mathbb{E}_{p_t(x)} \left[\frac{1}{2\sigma_t^2} \|u_t(x) - f_t(x)\|^2 - \nabla s_t(x) \cdot u_t(x) - \partial_t s_t(x) - \frac{\sigma_t^2}{2} \Delta s_t(s) \right] \mathrm{d}t$$

Notice that only a few terms depend on the drift u_t. Thus to find the optimal drift u_t^* we will minimize:

$$J[u] = \int_0^1 \int_{\mathbb{R}^d} \left[\frac{1}{2\sigma_t^2} \|u_t(x) - f_t(x)\|^2 - \nabla s_t(x) \cdot u_t(x) \right] p_t(x) \mathrm{d}x \mathrm{d}t$$

Let $\eta_t(x)$ be an arbitrary smooth, compactly supported vector field and let $\epsilon \in \mathbb{R}$ be a scalar pertubation. We evaluate the functional at the perturbed vector field:

$$J[u + \epsilon\eta] = \int_0^1 \int_{\mathbb{R}^d} \left[\frac{1}{2\sigma_t^2} \|u_t(x) + \epsilon\eta_t(x) - f_t(x)\|^2 - \nabla s_t(x) \cdot (u_t(x) + \epsilon\eta_t(x)) \right] p_t(x) \mathrm{d}x \mathrm{d}t$$

We expand the norm:

$$\|u_t(x) + \epsilon\eta_t(x) - f_t(x)\|^2 = \|u_t(x) - f_t(x)\|^2 + 2\epsilon\eta_t(x) \cdot (u_t(x) - f_t(x)) + \epsilon^2 \|\eta_t(x)\|^2$$

And for the second term:

$$\nabla s_t(x) \cdot (u_t(x) + \epsilon \eta_t(x)) = \nabla s_t(x) \cdot u_t(x) + \epsilon \, \nabla s_t(x) \cdot \eta_t(x)$$

To find the minimum we take the first variation of J with respect to ϵ evaluated at $\epsilon = 0$ and set it to zero:

$$\frac{\mathrm{d}}{\mathrm{d}\epsilon} J[u + \epsilon \eta] \mid_{\epsilon=0} = 0$$

Thus we set:

$$\int_0^1 \int_{\mathbb{R}^d} \left[\frac{1}{\sigma_t^2} \left(u_t(x) - f_t(x) \right) \cdot \eta_t(x) - \nabla s_t(x) \cdot \eta_t(x) \right] p_t(x) \mathrm{d}x \mathrm{d}t = 0$$

Then, we factor the arbitrary test function:

$$\int_0^1 \int_{\mathbb{R}^d} p_t(x) \left[\eta_t(x) \cdot \left(\frac{1}{\sigma_t^2} \left(u_t(x) - f_t(x) \right) - \nabla s_t(x) \right) \right] \mathrm{d}x \mathrm{d}t = 0$$

By the fundamental lemma of calculus of variations, because $\eta_t(x)$ is arbitrary:

$$p_t(x) \left[\frac{1}{\sigma_t^2} (u_t(x) - f_t(x)) - \nabla s_t(x) \right] = 0$$

Assuming that $p_t(x) > 0$ inside the support of the distribution:

$$\frac{1}{\sigma_t^2} (u_t(x) - f_t(x)) - \nabla s_t(x) = 0$$

Thus the optimal drift takes the following form:

$$u_t^*(x) = f_t(x) + \sigma_t^2 \nabla s_t(x)$$

Thus we can eliminate u_t by substituting the experssion of the optimal drift back into the Langrangian:

$$\begin{aligned} \frac{1}{2\sigma_t^2} \| u_t^* - f_t \|^2 - \nabla s_t \cdot u_t^* &= \frac{1}{2\sigma_t^2} \| (f_t + \sigma_t^2 \nabla s_t) - f_t \|^2 - \nabla s_t \cdot (f + \sigma_t^2 \nabla s_t) \\ &= -\frac{\sigma_t^2}{2} \| \nabla s_t \|^2 - \nabla s_t \cdot f_t \end{aligned}$$

Thus the unconstraint objective becomes:

$$\begin{aligned} \mathcal{K}_{\mathrm{SB}} = \sup_{s_t} \Bigg\{ & \mathbb{E}_{p_1(x)}[s_1] - \mathbb{E}_{p_0(x)}[s_0] \\ & - \int_0^1 \mathbb{E}_{p_t(x)} \left[\frac{\sigma_t^2}{2} \| \nabla s_t \|^2 + \nabla s_t \cdot f_t + \partial_t s_t + \frac{\sigma_t^2}{2} \Delta s_t \right] \mathrm{d}t \Bigg\} \end{aligned}$$

Finally, we turn this maximization objective into a minimization by minimizing the negative functional and derive the loss of the SB problem:

$$L_{\text{SB}} = \inf_{s_t} \Bigg\{ \mathbb{E}_{p_0(x)}[s_0(x)] - \mathbb{E}_{p_1(x)}[s_1(x)] + \int_0^1 \mathbb{E}_{p_t(x)} \left[\frac{\sigma_t^2}{2} \|\nabla s_t(x)\|^2 + \nabla s_t(x) \cdot f_t(x) + \partial_t s_t(x) + \frac{\sigma_t^2}{2} \Delta s_t(x) \right] \mathrm{d}t \Bigg\}$$

3. Iterative Action Matching Solver

Conditional action matching objective

This section details the application of the previously derived Action Matching objective to the marginal Schrödinger Bridge (SB) problem. Specifically, we address the setting where the intermediate path evolution is unknown, and only independent samples from the boundary marginal distributions, p_0 and p_1, are accessible.

To construct a tractable iterative solver that bypasses boundary singularities while utilizing exact sampling, we restrict the reference measure $R \in \mathcal{P}(\Omega)$ to the family of processes governed by linear stochastic differential equations (SDEs):

$$\mathrm{d}X_t = (\alpha(t)X_t + \beta(t))\mathrm{d}t + \sigma(t)\mathrm{d}W_t$$

where $\alpha : [0,1] \to \mathbb{R}$, $\beta : [0,1] \to \mathbb{R}^d$, and $\sigma : [0,1] \to (0,\infty)$ are deterministic, continuous functions. A first analysis of those processes can be found here.

Let $P \in \mathcal{P}(\Omega)$ denote the optimal path measure solving the Schrödinger Bridge problem with respect to the reference measure R and the prescribed boundary marginals p_0 and p_1. It is a well-established property that P belongs to the reciprocal class of the reference measure, denoted as $P \in \mathcal{R}(R)$. Consequently, the optimal measure P and the reference measure R share the identical Markovian bridges:

$$P(\cdot \mid X_0 = x, X_1 = y) = R(\cdot \mid X_0 = x, X_1 = y)$$

By projecting these path measures onto the state space at an intermediate time $t \in (0,1)$, we obtain the equivalence of their one-time conditional distributions:

$$P_t(\cdot \mid X_0 = x_0, X_1 = x_1) = R_t(\cdot \mid X_0 = x_0, X_1 = x_1)$$

Assuming that the finite-dimensional distributions P_t and R_t are absolutely continuous with respect to the Lebesgue measure on $\mathbb{R}^d$, we denote their respective conditional

probability density functions as $p_{t|01}$ and $q_{t|01}$. This yields the exact equivalence of the conditional bridge densities:

$$p_{t|01}(x_t \mid x_0, x_1) = q_{t|01}(x_t \mid x_0, x_1), \qquad \forall x_0, x_1, x_t \in \mathbb{R}^d$$

By the law of total probability, we disintegrate the intermediate marginal density $p_t(x_t)$ over the optimal static coupling $\pi^* \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$ of the boundary conditions:

$$\begin{aligned} p_t(x_t) &= \int_{\mathbb{R}^d \times \mathbb{R}^d} q_{t|01}(x_t \mid x_0, x_1) \mathrm{d}\pi(x_0, x_1) \\ &= \mathbb{E}_{(X_0, X_1) \sim \pi}[q_{t|01}(x_t \mid X_0, X_1)] \end{aligned}$$

Consequently, for any integrable function, the expectation under the intractable marginal density p_t translates exactly into a nested expectation over the optimal joint boundaries and the tractable reference bridge:

$$\mathbb{E}_{X_t \sim p_t}[\,\cdot\,] = \mathbb{E}_{(X_0, X_1) \sim \pi}[\mathbb{E}_{X_t \sim q_{t|01}(\cdot | X_0, X_1)}[\,\cdot\,]]$$

By substituting this disintegrated expectation back into the continuous-time Schrödinger Bridge loss function, and expressing the time integral as an expectation over $t \sim U[0,1]$, we derive the conditional Entropic Action Matching objective:

$$\begin{aligned} L_{\mathrm{SB}} = \inf_{s_t} \Bigg\{ & \mathbb{E}_{x_0 \sim p_0}[s_0(x_0)] - \mathbb{E}_{x_1 \sim p_1}[s_1(x_1)] \\ & + \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{(x_0, x_1) \sim \pi,\, x_t \sim q_{t|01}(\cdot | x_0, x_1)} \left[\frac{\sigma_t^2}{2} \|\nabla s_t(x_t)\|^2 + \nabla s_t(x_t) \cdot f_t(x_t) + \partial_t s_t(x_t) + \frac{\sigma_t^2}{2} \Delta s_t(x_t) \right] \Bigg\} \end{aligned}$$

To evaluate the nested expectation, we require the closed-form expression for the reference conditional bridge density, $q_{t|01}(x_t \mid x_0, x_1)$. Because the reference measure is governed by a linear SDE with deterministic coefficients, all finite-dimensional conditional distributions of the state are strictly Gaussian:

$$q_{t|01}(x_t \mid x_0, x_1) = \mathcal{N}(x_t\,; \mu_t(x_0, x_1), \Sigma_t)$$

Drawing upon the prior analysis, for any times $s < t$, the forward transition kernel $q_{st}(x_t \mid x_s)$ is Gaussian with its mean and covariance given by:

$$\begin{aligned} \mu_{st}(x) &= \mathbb{E}[X_t \mid X_s = x] = A_{st} x + B_{st} \\ \Sigma_{st} &= \mathrm{Cov}[X_t \mid X_s = x] = V_{st} I_d \end{aligned}$$

where the deterministic scaling factor $A_{st} \in \mathbb{R}$, the drift vector $B_{st} \in \mathbb{R}^d$, and the variance scalar $V_{st} \in \mathbb{R}^+$ are defined exactly as:

$$\begin{aligned} A_{st} &= \exp\left(\int_s^t \alpha(u)\mathrm{d}u\right) \\ B_{st} &= \int_s^t A_{\tau t}\beta(\tau)\mathrm{d}\tau \\ V_{st} &= \int_s^t A_{\tau t}^2\sigma^2(\tau)\mathrm{d}\tau \end{aligned}$$

Our objective is to derive the exact expression for the conditional bridge density $q_{t|01}(x_t \mid x_0, x_1)$. By the definition of conditional probability, we can express this density in terms of the forward transition kernels:

$$\begin{aligned} q_{t|01}(x_t \mid x_0, x_1) &= \frac{q(x_t, x_1 \mid x_0)}{q_{01}(x_1 \mid x_0)} \\ &\propto q(x_t, x_1 \mid x_0) \end{aligned}$$

Applying the chain rule of probability to the joint conditional density yields:

$$q(x_t, x_1 \mid x_0) = q_{0t}(x_t \mid x_0)q(x_1 \mid x_t, x_0)$$

Because the reference measure R satisfies the Markov property, the future state X_1 is conditionally independent of the past state X_0 given the present state X_t. Therefore, $q(x_1 \mid x_t, x_0) = q_{t1}(x_1 \mid x_t)$. Substituting this result establishes that the conditional bridge density is strictly proportional to the product of the forward transitions:

$$q_{t|01}(x_t \mid x_0, x_1) \propto q_{0t}(x_t \mid x_0)q_{t1}(x_1 \mid x_t)$$

As established, both forward transition kernels $q_{0t}(x_t \mid x_0)$ and $q_{t1}(x_1 \mid x_t)$ are exact, closed-form Gaussian distributions:

$$\begin{aligned} q_{0t}(x_t \mid x_0) &= \mathcal{N}(x_t;\ \mu_{0t}(x_0), V_{0t}I_d) \\ q_{t1}(x_1 \mid x_t) &= \mathcal{N}(x_1;\ A_{t1}x_t + B_{t1}, V_{t1}I_d) \end{aligned}$$

where their respective covariance matrices are defined as $\Sigma_{0t} = V_{0t}I_d$ and $\Sigma_{t1} = V_{t1}I_d$.

By substituting these densities into the proportionality relation and isolating the terms in the exponent dependent on x_t, we obtain the unnormalized log-density:

$$\begin{aligned} q_{t|01}(x_t \mid x_0, x_1) \propto \exp\Bigg[-\frac{1}{2}\Bigg(&(x_t - \mu_{0t}(x_0))^\top \Sigma_{0t}^{-1}(x_t - \mu_{0t}(x_0)) \\ &+ (A_{t1}x_t + B_{t1} - x_1)^\top \Sigma_{t1}^{-1}(A_{t1}x_t + B_{t1} - x_1)\Bigg)\Bigg] \end{aligned}$$

Because the product of two Gaussian probability density functions yields another unnormalized Gaussian, we can derive the exact conditional bridge parameters by completing the square with respect to x_t. Expanding the quadratic forms and matching the coefficients yields the exact conditional bridge covariance $\Sigma_(t)$ and mean $\mu_t(x_0, x_1)$:

$$\Sigma_t = \left(\Sigma_{0t}^{-1} + A_{t1}^2 \Sigma_{t1}^{-1}\right)^{-1}$$
$$\mu_t(x_0, x_1) = \Sigma_t \left(\Sigma_{0t}^{-1} \mu_{0t}(x_0) + A_{t1} \Sigma_{t1}^{-1}(x_1 - B_{t1})\right)$$

Finally, we can formulate the conditional Entropic Action Matching objective as follows:

$$L_{\text{SB}} = \inf_{s_t} \Bigg\{ \mathbb{E}_{x_0 \sim p_0}[s_0(x_0)] - \mathbb{E}_{x_1 \sim p_1}[s_1(x_1)] + \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{(x_0, x_1) \sim \pi,\, x_t \sim q_{t|01}(\cdot | x_0, x_1)} \left[\frac{\sigma_t^2}{2} \|\nabla s_t(x_t)\|^2 + \nabla s_t(x_t) \cdot f_t(x_t) + \partial_t s_t(x_t) + \frac{\sigma_t^2}{2} \Delta s_t(x_t) \right] \Bigg\}$$

where the reference drift is defined as $f_t(x_t) = \alpha(t) x_t + \beta(t)$, the volatility is $\sigma_t = \sigma(t)$, and the intermediate states are sampled exactly from the derived Gaussian bridge, $x_t \sim \mathcal{N}(\mu_t(x_0, x_1), \Sigma_t)$.

Backward action matching objective

Notice that the previously derived conditional action matching objective assumes that both the reference and the controlled processes evolve forward in time. Under this assumption, the optimal controlled drift is given by:

$$u_t^*(x_t) = f_t(x_t) + \sigma_t^2 \nabla s_t(x_t)$$

For the iterative algorithm—which we will detail later—we must also solve the Schrödinger Bridge problem in the reverse time direction. This requires training an action $s_t(x)$ associated with a backward-in-time stochastic differential equation. To derive this, we follow a path analogous to Section 2.

Let R^B denote the time-reversal of the reference path measure R. Let $q_t(x)$ denote the continuous time-marginal density of the forward reference process R with respect to the Lebesgue measure on $\mathbb{R}^d$. To rigorously define the backward dynamics without ambiguity in the differentials, we introduce the positive reverse time variable $\tau = 1 - t \in [0, 1]$, such that $\mathrm{d}\tau = -\mathrm{d}t > 0$. By Anderson's Theorem, the time-reversed process evaluated under this forward-flowing clock τ is governed by the following backward SDE:

$$R^B : \mathrm{d}X_{1-\tau} = -[f_{1-\tau}(X_{1-\tau}) - \sigma_{1-\tau}^2 \nabla \log q_{1-\tau}(X_{1-\tau})]\mathrm{d}\tau + \sigma_{1-\tau} \mathrm{d}\bar{W}_\tau^R$$

where $(\bar{W}_\tau^R)_{\tau\in[0,1]}$ is a standard Brownian motion adapted to the backward filtration under R. For notational convenience, we will express the states and parameters evaluated at physical time $t = 1 - \tau$. Defining the backward reference drift as $f_t^B(x) = f_t(x) - \sigma_t^2 \nabla \log q_t(x)$, we obtain the cleaner representation:

$$R^B : \mathrm{d}X_t = -f_t^B(X_t)\mathrm{d}\tau + \sigma_t \mathrm{d}\bar{W}_\tau^R$$

Similarly, let $P \in \mathcal{P}(\Omega)$ denote a controlled measure. Its time-reversal, denoted as P^B, governs the canonical process evolving backward in physical time according to the SDE:

$$P^B : \mathrm{d}X_t = -u_t^B(X_t)\mathrm{d}\tau + \sigma_t \mathrm{d}\bar{W}_\tau^P$$

where $u_t^B : [0,1] \times \mathbb{R}^d \to \mathbb{R}^d$ is the backward control drift parameterized by physical time t, and $(\bar{W}_\tau^P)_{\tau\in[0,1]}$ is a standard Brownian motion adapted to the backward filtration under P^B. Note that the negative sign on the drifts rigorously accounts for the state trajectory being traversed in reverse over the positive increment $\mathrm{d}\tau$.

By isolating the Brownian motion under R^B and substituting the dynamics of P^B, we derive the relationship between the noise processes:

$$\begin{aligned}
\sigma_t \mathrm{d}\bar{W}_\tau^R &= \mathrm{d}X_t + f_t^B(X_t)\mathrm{d}\tau \\
&= (-u_t^B(X_t)\mathrm{d}\tau + \sigma_t \mathrm{d}\bar{W}_\tau^P) + f_t^B(X_t)\mathrm{d}\tau \\
\mathrm{d}\bar{W}_\tau^R &= \mathrm{d}\bar{W}_\tau^P + \frac{1}{\sigma_t}\left(f_t^B(X_t) - u_t^B(X_t)\right)\mathrm{d}\tau
\end{aligned}$$

We define the scaled relative drift evaluated at physical time t as $v_t^B(x) = \frac{1}{\sigma_t}(u_t^B(x) - f_t^B(x))$. Substituting this yields:

$$\mathrm{d}\bar{W}_\tau^R = \mathrm{d}\bar{W}_\tau^P - v_t^B(X_t)\mathrm{d}\tau$$

Because we have formulated the relative drift with respect to the positive backward increment $\mathrm{d}\tau > 0$, Girsanov's theorem is rigorously well-defined without any ambiguity regarding negative variances. The density of P^B with respect to R^B over the path is given by the standard stochastic exponential martingale integrated over $\tau \in [0,1]$:

$$\frac{\mathrm{d}P^B}{\mathrm{d}R^B} = \exp\left(\int_0^1 -v_{1-\tau}^B(X_{1-\tau}) \cdot \mathrm{d}\bar{W}_\tau^R - \frac{1}{2}\int_0^1 \| - v_{1-\tau}^B(X_{1-\tau})\|^2 \mathrm{d}\tau\right)$$

Consequently, the KL divergence between the backward measures is strictly given by the expectation of the quadratic variation over $\tau \in [0,1]$. Since the squared norm eliminates the inner negative sign ($\| - v_t^B\|^2 = \|v_t^B\|^2$), we have:

$$\mathrm{KL}(P^B \mid R^B) = \frac{1}{2}\,\mathbb{E}_{P^B}\left[\int_0^1 \|v_{1-\tau}^B(X_{1-\tau})\|^2 \mathrm{d}\tau\right]$$

Finally, having rigorously established the KL divergence, we can perform a simple change of variables back to the physical time domain. By substituting $\tau = 1 - t$, the integration limits map as $0 \to 1$ and $1 \to 0$, and the differential maps as $\mathrm{d}\tau = -\mathrm{d}t$. Reversing the integral bounds absorbs the negative sign ($\int_1^0 -\mathrm{d}t = \int_0^1 \mathrm{d}t$), yielding the KL divergence completely in terms of physical time $t \in [0, 1]$:

$$\begin{aligned}\mathrm{KL}(P^B \mid R^B) &= \frac{1}{2}\,\mathbb{E}_{P^B}\left[\int_0^1 \|v_t^B(X_t)\|^2 \mathrm{d}t\right] \\ &= \int_0^1 \frac{1}{2\sigma_t^2}\,\mathbb{E}_{x\sim p_t}[\|u_t^B(x) - f_t^B(x)\|^2]\mathrm{d}t\end{aligned}$$

where $p_t(x)$ is the continuous time-marginal density of the controlled process P.

Furthermore, we must derive the dynamics of this density to constrain the optimization. To avoid the intricacies of backward Itô calculus, we analyze the density evolution directly in the physical forward time t. The forward-evolving density $p_t(x)$ satisfies the standard Fokker-Planck equation governed by the forward drift $u_t(x)$:

$$\partial_t p_t(x) = -\nabla \cdot (p_t(x) u_t(x)) + \frac{\sigma_t^2}{2}\Delta p_t(x)$$

By Nelson's kinematics, the forward and backward drifts are strictly related by $u_t(x) = u_t^B(x) + \sigma_t^2 \nabla \log p_t(x)$. Substituting this relationship into the forward equation, and noting that $p_t(x)\nabla \log p_t(x) = \nabla p_t(x)$, we obtain:

$$\begin{aligned}\partial_t p_t(x) &= -\nabla \cdot \Big(p_t(x) u_t^B(x) + \sigma_t^2 \nabla p_t(x)\Big) + \frac{\sigma_t^2}{2}\Delta p_t(x) \\ &= -\nabla \cdot (p_t(x) u_t^B(x)) - \sigma_t^2 \Delta p_t(x) + \frac{\sigma_t^2}{2}\Delta p_t(x) \\ &= -\nabla \cdot (p_t(x) u_t^B(x)) - \frac{\sigma_t^2}{2}\Delta p_t(x)\end{aligned}$$

Therefore, we can formulate the backward Schrödinger Bridge optimization problem entirely in the physical forward time $t \in [0, 1]$ as:

$$\begin{aligned}\mathcal{K}_{\mathrm{SB}}^B = \inf_{u_t^B} \left\{ \int_0^1 \mathbb{E}_{x_t\sim p_t}\left[\frac{1}{2\sigma_t^2}\,\|u_t^B(x_t) - f_t^B(x_t)\|^2\right] \mathrm{d}t \right\} \quad \text{s.t.} \\ \partial_t p_t(x) = -\nabla \cdot (p_t(x) u_t^B(x)) - \frac{\sigma_t^2}{2}\Delta p_t(x)\end{aligned}$$

We introduce a Lagrangian multiplier function $s_t^B(x)$ to enforce the physical-time Fokker-Planck constraint. Integrating this constraint over the forward time domain $t \in [0, 1]$ establishes the boundary densities unambiguously as $p_{t=0} = p_0$ and $p_{t=1} = p_1$.

Applying integration by parts to the spatial and temporal derivatives maps the constraint directly into the objective function:

$$\mathcal{K}^B_{\mathrm{SB}} = \sup_{s_t^B} \inf_{u_t^B} \Bigg\{ \mathbb{E}_{x_1 \sim p_1}[s_1^B(x_1)] - \mathbb{E}_{x_0 \sim p_0}[s_0^B(x_0)] + \int_0^1 \mathbb{E}_{x_t \sim p_t} \left[\frac{1}{2\sigma_t^2} \| u_t^B(x_t) - f_t^B(x_t) \|^2 - \nabla s_t^B(x_t) \cdot u_t^B(x_t) - \partial_t s_t^B(x_t) + \frac{\sigma_t^2}{2} \Delta s_t^B(x_t) \right] \mathrm{d}t \Bigg\}$$

The minimizer is extracted by completing the square for all terms involving the control drift, defining the exact optimal backward drift:

$$u_t^{B*}(x) = f_t^B(x) + \sigma_t^2 \nabla s_t^B(x)$$

By substituting this optimal drift back into the objective and negating the functional to convert the supremum into an infimum, we derive the exact backward conditional Entropic Action Matching objective:

$$L^B_{\mathrm{SB}} = \inf_{s_t^B} \Bigg\{ \mathbb{E}_{x_0 \sim p_0}[s_0^B(x_0)] - \mathbb{E}_{x_1 \sim p_1}[s_1^B(x_1)] + \int_0^1 \mathbb{E}_{x_t \sim p_t} \left[\frac{\sigma_t^2}{2} \| \nabla s_t^B(x_t) \|^2 + \nabla s_t^B(x_t) \cdot f_t^B(x_t) + \partial_t s_t^B(x_t) - \frac{\sigma_t^2}{2} \Delta s_t^B(x_t) \right] \mathrm{d}t \Bigg\}$$

Crafting the algorithm

We have derived both the forward and backward action matching objectives. If the marginal density p_t was available via samples, these objectives could be used directly to solve the Schrödinger Bridge problem. Unfortunately, in our setting, we only have access to independent samples from the boundary marginals p_0 and p_1.

To optimize these objectives in practice, we must approximate the continuous time integrals using Monte Carlo estimation. Because we rigorously anchored both the forward and backward objectives in the physical forward time $t \in [0, 1]$, we can seamlessly evaluate both integrals by drawing random times $t \sim U[0, 1]$.

We now utilize the conditional form of the action matching objectives. Following the derivation at the beginning of Section 3, the expectation under the intractable marginal density p_t can be expressed as a nested expectation over the empirical joint boundaries and the tractable reference bridge. For the backward objective, we disintegrate p_t over the backward empirical coupling $\pi^B(x_0, x_1)$ (which connects the target distribution p_1 back to p_0):

$$\mathbb{E}_{x_t \sim p_t}[\,\cdot\,] = \mathbb{E}_{(x_0, x_1) \sim \pi^B}[\mathbb{E}_{x_t \sim q_{t|01}(\cdot|x_0, x_1)}[\,\cdot\,]]$$

Crucially, because the reference process R and its time-reversal R^B induce the exact same distribution over geometric paths, the conditional bridge density evaluated at physical time t is identical regardless of the direction of the optimization. This allows us to simulate the intermediate states for both the forward and backward objectives using the exact same closed-form Gaussian bridge parameters derived previously:

$$q_{t|01}(x_t \mid x_0, x_1) = \mathcal{N}(x_t;\ \mu_t(x_0, x_1), \Sigma_t)$$

By substituting this disintegrated expectation into both the forward and backward loss functions, we formulate the final, tractable conditional Schrödinger Bridge action matching objectives:

$$L_{\mathrm{SB}}^F = \inf_{s_t^F} \Bigg\{ \mathbb{E}_{x_0\sim p_0}[s_0^F(x_0)] - \mathbb{E}_{x_1\sim p_1}[s_1^F(x_1)] + \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{(x_0,x_1)\sim\pi^F,\, x_t\sim q_{t|01}(\cdot|x_0,x_1)} \left[\frac{\sigma_t^2}{2}\|\nabla s_t^F(x_t)\|^2 + \nabla s_t^F(x_t)\cdot f_t(x_t) + \partial_t s_t^F(x_t) + \frac{\sigma_t^2}{2}\Delta s_t^F(x_t) \right] \Bigg\}$$

$$L_{\mathrm{SB}}^B = \inf_{s_t^B} \Bigg\{ \mathbb{E}_{x_0\sim p_0}[s_0^B(x_0)] - \mathbb{E}_{x_1\sim p_1}[s_1^B(x_1)] + \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{(x_0,x_1)\sim\pi^B,\, x_t\sim q_{t|01}(\cdot|x_0,x_1)} \left[\frac{\sigma_t^2}{2}\,\|\nabla s_t^B(x_t)\|^2 + \nabla s_t^B(x_t)\cdot f_t^B(x_t) + \partial_t s_t^B(x_t) - \frac{\sigma_t^2}{2}\Delta s_t^B(x_t) \right] \Bigg\}$$

A crucial detail in evaluating these objectives lies in the relationship between the boundary marginal distributions (p_0, p_1) and the empirical joint couplings (π^F, π^B). Because the boundary expectations $\mathbb{E}_{p_0}$ and $\mathbb{E}_{p_1}$ were derived via integration by parts over the time-marginal density p_t, they must strictly correspond to the exact first and second marginals of the coupling used to disintegrate p_t.

By grouping these boundary terms under a single expectation over the joint couplings, we arrive at our final, fully tractable formulation for the iterative action matching solver:

$$L_{\mathrm{SB}}^F = \inf_{s_t^F} \Bigg\{ \mathbb{E}_{(x_0,x_1)\sim\pi^F}[s_0^F(x_0) - s_1^F(x_1)] + \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{(x_0,x_1)\sim\pi^F,\, x_t\sim q_{t|01}(\cdot|x_0,x_1)} \left[\frac{\sigma_t^2}{2}\|\nabla s_t^F(x_t)\|^2 + \nabla s_t^F(x_t)\cdot f_t(x_t) + \partial_t s_t^F(x_t) + \frac{\sigma_t^2}{2}\Delta s_t^F(x_t) \right] \Bigg\}$$

$$L^B_{\mathrm{SB}} = \inf_{s^B_t} \Bigg\{ \mathbb{E}_{(x_0,x_1)\sim\pi^B}[s^B_0(x_0) - s^B_1(x_1)]$$
$$+ \mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{(x_0,x_1)\sim\pi^B,\, x_t\sim q_{t|01}(\cdot|x_0,x_1)} \left[\frac{\sigma_t^2}{2} \|\nabla s^B_t(x_t)\|^2 + \nabla s^B_t(x_t)\cdot f^B_t(x_t) + \partial_t s^B_t(x_t) - \frac{\sigma_t^2}{2}\Delta s^B_t(x_t) \right] \Bigg\}$$

A significant computational bottleneck in evaluating the backward objective L^B_{SB} lies in the backward reference drift $f^B_t(x) = f_t(x) - \sigma_t^2 \nabla \log q_t(x)$. Because $q_t(x)$ is the continuous time-marginal of the reference process, its score function $\nabla \log q_t(x)$ is generally intractable.

However, a profound simplification occurs if we reparameterize the backward Lagrangian multiplier. Note that the optimal backward drift is given by $u^{B*}_t(x) = f^B_t(x) + \sigma_t^2 \nabla s^B_t(x)$. By substituting $f^B_t(x)$, we obtain:

$$u^{B*}_t(x) = f_t(x) + \sigma_t^2 \nabla \big(s^B_t(x) - \log q_t(x) \big)$$

Let us define a combined backward action potential $\tilde{s}^B_t(x) := s^B_t(x) - \log q_t(x)$. The optimal backward drift simplifies strictly to $u^{B*}_t(x) = f_t(x) + \sigma_t^2 \nabla \tilde{s}^B_t(x)$, relying only on the known forward reference drift $f_t(x)$.

To determine the objective that $\tilde{s}^B_t(x)$ minimizes, we substitute $s^B_t(x) = \tilde{s}^B_t(x) + \log q_t(x)$ into the integrand of L^B_{SB}:

$$\begin{aligned}
&\frac{\sigma_t^2}{2} \|\nabla s^B_t(x_t)\|^2 + \nabla s^B_t(x_t)\cdot f^B_t(x_t) + \partial_t s^B_t(x_t) - \frac{\sigma_t^2}{2}\Delta s^B_t(x_t) \\
&\quad = \frac{\sigma_t^2}{2} \|\nabla \tilde{s}^B_t(x_t) + \nabla \log q_t(x_t)\|^2 + \nabla \tilde{s}^B_t(x_t)\cdot f^B_t(x_t) + \nabla \log q_t(x_t)\cdot f^B_t(x_t) \\
&\qquad + \partial_t \tilde{s}^B_t(x_t) + \partial_t \log q_t(x_t) - \frac{\sigma_t^2}{2}\Delta \tilde{s}^B_t(x_t) - \frac{\sigma_t^2}{2}\Delta \log q_t(x_t)
\end{aligned}$$

By substituting the backward reference drift $f^B_t(x) = f_t(x) - \sigma_t^2 \nabla \log q_t(x)$ we derive:

$$\begin{aligned}
&\frac{\sigma_t^2}{2} \|\nabla s^B_t(x_t)\|^2 + \nabla s^B_t(x_t)\cdot f^B_t(x_t) + \partial_t s^B_t(x_t) - \frac{\sigma_t^2}{2}\Delta s^B_t(x_t) \\
&\quad = \frac{\sigma_t^2}{2} \|\nabla \tilde{s}^B_t(x_t) + \nabla \log q_t(x_t)\|^2 + \nabla \tilde{s}^B_t(x_t)\cdot f_t(x_t) - \sigma_t^2 \nabla \tilde{s}^B_t(x_t)\cdot \nabla \log q_t(x_t) \\
&\qquad + \nabla \log q_t(x_t)\cdot f_t(x_t) - \sigma_t^2 \nabla \log q_t(x_t)\cdot \nabla \log q_t(x_t) \\
&\qquad + \partial_t \tilde{s}^B_t(x_t) + \partial_t \log q_t(x_t) - \frac{\sigma_t^2}{2}\Delta \tilde{s}^B_t(x_t) - \frac{\sigma_t^2}{2}\Delta \log q_t(x_t)
\end{aligned}$$

Expanding the squared norm and dot products reveals an exact algebraic cancellation of the cross-terms involving $\nabla \tilde{s}_t^B \cdot \nabla \log q_t$:

$$\begin{aligned}
&\frac{\sigma_t^2}{2} \|\nabla s_t^B(x_t)\|^2 + \nabla s_t^B(x_t) \cdot f_t^B(x_t) + \partial_t s_t^B(x_t) - \frac{\sigma_t^2}{2} \Delta s_t^B(x_t) \\
&\quad = \frac{\sigma_t^2}{2} \|\nabla \tilde{s}_t^B(x_t)\|^2 + \frac{\sigma_t^2}{2} \|\nabla \log q_t(x_t)\|^2 + \nabla \tilde{s}_t^B(x_t) \cdot f_t(x_t) \\
&\qquad + \nabla \log q_t(x_t) \cdot f_t(x_t) - \sigma_t^2 \nabla \log q_t(x_t) \cdot \nabla \log q_t(x_t) \\
&\qquad + \partial_t \tilde{s}_t^B(x_t) + \partial_t \log q_t(x_t) - \frac{\sigma_t^2}{2} \Delta \tilde{s}_t^B(x_t) - \frac{\sigma_t^2}{2} \Delta \log q_t(x_t)
\end{aligned}$$

Because Action Matching optimizes the vector field over a fixed batch of sampled paths from the reference bridge, all remaining terms containing only $\log q_t(x)$ act as constants with respect to the optimization of $\tilde{s}_t^B$:

$$\begin{aligned}
&\frac{\sigma_t^2}{2} \|\nabla s_t^B(x_t)\|^2 + \nabla s_t^B(x_t) \cdot f_t^B(x_t) + \partial_t s_t^B(x_t) - \frac{\sigma_t^2}{2} \Delta s_t^B(x_t) \\
&\quad \propto \frac{\sigma_t^2}{2} \|\nabla \tilde{s}_t^B(x_t)\|^2 + \nabla \tilde{s}_t^B(x_t) \cdot f_t(x_t) + \partial_t \tilde{s}_t^B(x_t) - \frac{\sigma_t^2}{2} \Delta \tilde{s}_t^B(x_t)
\end{aligned}$$

Consequently, the backward objective structurally collapses into a similar form as the forward objective, replacing s_t^B with $\tilde{s}_t^B$ and f_t^B with f_t:

$$\begin{aligned}
L_{\mathrm{SB}}^B = \inf_{\tilde{s}_t^B} \Bigg\{ &\mathbb{E}_{(x_0,x_1)\sim\pi^B}[\tilde{s}_0^B(x_0) - \tilde{s}_1^B(x_1)] \\
&+ \mathbb{E}_{t\sim U[0,1]} \mathbb{E}_{(x_0,x_1)\sim\pi^B,\, x_t\sim q_{t|01}(\cdot|x_0,x_1)} \left[\frac{\sigma_t^2}{2} \|\nabla \tilde{s}_t^B(x_t)\|^2 + \nabla \tilde{s}_t^B(x_t) \cdot f_t(x_t) + \partial_t \tilde{s}_t^B(x_t) - \frac{\sigma_t^2}{2} \Delta \tilde{s}_t^B(x_t) \right] \Bigg\}
\end{aligned}$$

By absorbing the intractable reference score into the backward potential, both the forward and backward optimization steps utilize the exact same analytical Gaussian bridge $q_{t|01}$ and the exact same forward reference drift $f_t(x)$. This yields a fully tractable, self-contained iterative algorithm.

We are ready to present the algorithm. We initialize the backward empirical coupling $\pi^{B,(0)}(x_0, x_1)$ by drawing independent samples from the boundary datasets $p_0(x_0)$ and $p_1(x_1)$ (or via a mini-batch optimal transport plan). Then begins the step 1: Backward Half step (k).

1. Train: Optimize the backward network $\tilde{s}_t^{(B,k)}$ by minimizing L_{SB}^B using boundary pairs $(x_0, x_1) \sim \pi^{(B,k)}$ and intermediate states sampled directly from the analytical bridge $x_t \sim q_{t|01}(\cdot \mid x_0, x_1)$.

2. Simulate: Generate the forward coupling $\pi^{(F,k)}$ by drawing target samples $x_1 \sim p_1$ and simulating the learned backward SDE from $t = 1$ down to $t = 0$:

$$\mathrm{d}X_t = -\big(f_t(X_t) + \sigma_t^2 \nabla \tilde{s}_t^{(B,k)}(X_t)\big)\mathrm{d}\tau + \sigma_t \mathrm{d}\bar{W}_\tau^P$$

The resulting simulated trajectories furnish the initial states X_0, forming the new empirical coupling pairs $(X_0, x_1) \sim \pi^{(F,k)}$.

Next the Forward Half step (k) begins.

1. Train: Optimize the forward network $s_t^{(F,k)}$ by minimizing L_{SB}^F using boundary pairs $(x_0, x_1) \sim \pi^{(F,k)}$ and intermediate states sampled directly from the analytical bridge $x_t \sim q_{t|01}(\cdot \mid x_0, x_1)$.

2. Simulate: Generate the backward coupling $\pi^{(B,k+1)}$ by drawing source samples $x_0 \sim p_0$ and simulating the learned forward SDE from $t = 0$ up to $t = 1$:

$$\mathrm{d}X_t = \big(f_t(X_t) + \sigma_t^2 \nabla s_t^{(F,k)}(X_t)\big)\mathrm{d}t + \sigma_t \mathrm{d}W_t^P$$

The resulting simulated trajectories furnish the terminal states X_1, forming the new empirical coupling pairs $(x_0, X_1) \sim \pi^{(B,k+1)}$.

We repeat the half steps until convergence.

Unlike standard IMF, which projects onto the Markovian class via the conditional expectation of the bridge drift, Action Matching restricts the drift to $f_t + \sigma_t^2 \nabla s_t$. This is the $L^2(p_t^\pi)$ Helmholtz–Hodge projection of the Markovian drift onto gradient fields, i.e. it removes the component w_t with $\nabla \cdot (p_t^\pi w_t) = 0$. The marginals $(p_t^\pi)_t$ are preserved, but when the Markovian drift is non-conservative (e.g. between Gaussians with non-commuting covariances) the path law and the resulting endpoint couplings differ from those of IMF. The projection is still orthogonal in the KL sense, and the SB drift $f_t + \sigma_t^2 \nabla \log \varphi_t$ lies in the gradient class. The SB is therefore a fixed point, and the monotone-decrease argument of IMF carries over to the alternating projections between $\mathcal{R}(R)$ and the gradient class, which we expect to yield convergence (a full proof is left to be done).

Algorithmic Implementations

The detailed alogrithm can be found in Algorithm 2. In this section we are going to discuss some of its implementations.

1.
Steps 1, and 3 required the first order derivites of the action ∇s_t, $\partial_t s_t$. Since the neural network maps the scalar time t and the d-dimensional state x to a scalar action value $s \in \mathbb{R}$, the first-order gradients are computationally cheap.

By feeding the concatenated input (t, x) into the network, a single backward pass (reverse-mode autodiff) yields both the spatial gradient $\nabla s_t(x) \in \mathbb{R}^d$ and the temporal derivative $\partial_t s_t(x) \in \mathbb{R}$ simultaneously. The computational cost of this is identical to a standard network training step: $\mathcal{O}(1)$ forward passes and $\mathcal{O}(1)$ backward passes.

The bottleneck is the Laplacian Δs_t. The Laplacian is the trace of the spatial Hessian matrix: $\Delta s_t(x) = \mathrm{Tr}(\nabla_x^2 s_t(x))$. Computing the exact Hessian trace naively requires computing the spatial gradient $\nabla s_t(x)$ and then differentiating each of its d components independently. This requires d separate backward passes. If the state space is high-dimensional (e.g., images or large biological systems), an $\mathcal{O}(d)$ scaling factor makes the exact Laplacian prohibitively expensive.

To bypass the $\mathcal{O}(d)$ bottleneck, the standard approach is to use a stochastic trace estimator, most commonly Hutchinson's Estimator. Instead of computing the exact trace, we project the Hessian onto a random noise vector. Let $z \in \mathbb{R}^d$ be a random vector drawn from a distribution with zero mean and unit covariance, such as the standard normal $z \sim \mathcal{N}(0, I_d)$ or Rademacher distribution $z \sim \mathrm{Unif}(\{-1, 1\}^d)$.

The unbiased estimator for the Laplacian is given by:

$$\Delta s_t(x) \approx \mathbb{E}_z \left[z^\top \nabla_x^2 s_t(x) z \right]$$

Because the Hessian is symmetric, you can compute this quadratic form highly efficiently using a directional derivative approach known as a Jacobian-Vector Product (JVP) or by utilizing a double backward pass (Vector-Jacobian Product).

Mathematically, this is implemented by first computing the spatial gradient vector $v = \nabla s_t(x)$, and then taking the dot product with the noise vector z before differentiating a second time:

$$z^\top \nabla_x^2 s_t(x) z = z^\top \nabla_x (v^\top z)$$

This reduces the cost of the Laplacian to just one additional forward/backward pass, changing the computational complexity from $\mathcal{O}(d)$ to $\mathcal{O}(1)$, regardless of the state dimension.

2.
To implement this minimization using stochastic gradient-based optimizers (such as Adam), we must approximate the continuous expectation using Monte Carlo estimation. Rather than evaluating the full dataset, we compute the loss over a mini-batch of N samples by independently sampling the time, the boundaries, and the intermediate states.

Specifically, for each index $i \in \{1, \ldots, N\}$ in the mini-batch, we perform the following exact sequential sampling steps:

- Time sampling: Draw a physical time $t_i \sim U[0,1]$.
- Boundary sampling: Draw a pair of boundary states from the current empirical coupling, $(x_{0,i}, x_{1,i}) \sim \pi^{B,(k)}$.
- Bridge sampling: Given the sampled time and boundaries, compute the exact Gaussian bridge mean and covariance, and draw the intermediate state $x_{t,i} \sim \mathcal{N}(\mu_{t_i}(x_{0,i}, x_{1,i}), \Sigma_{t_i})$.

By substituting these discrete samples—and assuming the neural network parameters are denoted by θ—the expectation is rigorously expressed as the following empirical loss function to be minimized:

$$\hat{L}^B_{\mathrm{SB}}(\theta) = \frac{1}{N}\sum_{i=1}^{N}\Bigg[\tilde{s}_\theta(0,x_{0,i}) - \tilde{s}_\theta(1,x_{1,i}) + \frac{\sigma_{t_i}^2}{2}\|\nabla_x \tilde{s}_\theta(t_i,x_{t,i})\|^2 + \nabla_x \tilde{s}_\theta(t_i,x_{t,i})\cdot f_{t_i}(x_{t,i}) + \partial_t \tilde{s}_\theta(t_i,x_{t,i}) - \frac{\sigma_{t_i}^2}{2}\Delta_x \tilde{s}_\theta(t_i,x_{t,i})\Bigg]$$

This empirical sum represents the exact tensor operation that will be passed into your automatic differentiation framework to compute the parameter gradients $\nabla_\theta \hat{L}^B_{\mathrm{SB}}(\theta)$ (equivalently for the forward loss).

3.

During Step 2 of the algorithm, we must generate the forward empirical coupling $\pi^{F,(k)}$ by simulating the learned backward SDE. This step effectively transports a batch of samples from the target distribution p_1 back to the source domain at $t = 0$, utilizing the optimal backward action network $\tilde{s}^{B,(k)}$ trained in Step 1. In practice, this continuous-time simulation is executed using a numerical integration scheme, most commonly the Euler-Maruyama method.

Because we explicitly formulated the backward SDE with respect to the positive reverse-time increment $\mathrm{d}\tau$, the numerical discretization avoids any ambiguity with negative time steps.

We first draw a batch of N independent target samples from the boundary dataset, $x_{1,i} \sim p_1$, for $i \in \{1,\dots,N\}$. We set the initial state of the backward simulation (which corresponds to physical time $t = 1$) as $X_{1,i} = x_{1,i}$.

We then discretize the integration interval $[0,1]$ into M uniformly spaced steps. The positive reverse-time step size is $\Delta\tau = 1/M$. We define a discrete sequence of physical evaluation times $t_m = 1 - m\Delta\tau$ for $m = 0, 1, \dots, M-1$.

For each discrete step m from 0 to $M-1$, we evaluate the forward reference drift f_{t_m} and the spatial gradient of the learned backward network $\nabla\tilde{s}^{B,(k)}$ at the current

physical time t_m and current state $X_{t_m,i}$. We draw a standard Gaussian noise vector $z_{m,i} \sim \mathcal{N}(0, I_d)$ to simulate the Brownian increment $\Delta\bar{W}_\tau^P \approx \sqrt{\Delta\tau} z_{m,i}$. The state is updated forward in simulation time τ (which is strictly backward in physical time t):

$$X_{t_{m+1},i} = X_{t_m,i} - \big(f_{t_m}(X_{t_m,i}) + \sigma_{t_m}^2 \nabla \tilde{s}_{t_m}^{B,(k)}(X_{t_m,i})\big)\Delta\tau + \sigma_{t_m}\sqrt{\Delta\tau} z_{m,i}$$

After executing M integration steps, the trajectories reach physical time $t_M = 0$. Let $X_{0,i}$ denote these final simulated states. We pair these generated starting states with their exact target anchors from the initialization step. The resulting collection of pairs $\{(X_{0,i}, x_{1,i})\}_{i=1}^N$ is cached in memory. This dataset forms the exact empirical forward coupling $\pi^{F,(k)}$ that will be sampled during Step 3 to train the forward network $s^{F,(k)}$.

We apply the same method without the confusion of time reversal when solving the forward SDE on Step 4.

4.

So far, our derivations have assumed a general reference measure governed by a linear SDE with deterministic coefficients. We now specialize the framework to two canonical reference families: (i) the scaled σ-Brownian Motion (R_σ), and (ii) the variance-preserving Ornstein-Uhlenbeck (VP-OU) process (R_γ). Their respective forward dynamics are governed by:

$$\begin{aligned} R_\sigma &: \mathrm{d}X_t = \sigma \mathrm{d}W_t^{R_\sigma} \\ R_\gamma &: \mathrm{d}X_t = -\gamma X_t \mathrm{d}t + \sqrt{2\gamma}\mathrm{d}W_t^{R_\gamma} \end{aligned}$$

Matching these dynamics to our general linear SDE yields the following piecewise definitions for the forward reference drift $f_t(x)$ and diffusion coefficient σ_t:

$$f_t(x) = \begin{cases} 0 & \text{under } R_\sigma \\ -\gamma x & \text{under } R_\gamma \end{cases}, \qquad \sigma_t = \begin{cases} \sigma & \text{under } R_\sigma \\ \sqrt{2\gamma} & \text{under } R_\gamma \end{cases}$$

Substituting these coefficients into the linear transition kernels yields the exact Gaussian bridge densities $q_{t|01}(x_t \mid x_0, x_1) = \mathcal{N}(x_t\,; \mu_t(x_0, x_1), \Sigma_t)$ for each reference process:

$$\mu_t(x_0, x_1) = \begin{cases} (1-t)x_0 + t x_1 & \text{under } R_\sigma \\ \dfrac{1 - e^{-2\gamma(1-t)}}{1 - e^{-2\gamma}}\, e^{-\gamma t} x_0 + \dfrac{1 - e^{-2\gamma t}}{1 - e^{-2\gamma}}\, e^{-\gamma(1-t)} x_1 & \text{under } R_\gamma \end{cases}$$

$$\Sigma_t = \begin{cases} \sigma^2 t(1-t) I_d & \text{under } R_\sigma \\ \dfrac{(1 - e^{-2\gamma t})(1 - e^{-2\gamma(1-t)})}{1 - e^{-2\gamma}}\, I_d & \text{under } R_\gamma \end{cases}$$

Applying these exact parameters to Algorithm 2 specializes the training objectives and sampling dynamics for each half-step as follows:

Step 1 (Train Backward Network $\tilde{s}^{B,(k)}$): Minimizing L^B_{SB} over the backward coupling $\pi^{B,(k)}$ and reference bridge $q_{t|01}$ specializes to:

$$\tilde{s}^{B,(k)} = \mathrm{argmin}_{\tilde{s}} \left\{ \mathbb{E}_{t,\pi^{B,(k)},q_{t|01}} \left[\tilde{s}_0(x_0) - \tilde{s}_1(x_1) + \mathcal{I}^B(\tilde{s}_t, x_t) \right] \right\}$$

where the continuous-time backward integrand $\mathcal{I}^B$ branches according to the chosen reference:

$$\mathcal{I}^B(\tilde{s}_t, x_t) = \begin{cases} \dfrac{\sigma^2}{2}\|\nabla \tilde{s}_t(x_t)\|^2 + \partial_t \tilde{s}_t(x_t) - \dfrac{\sigma^2}{2}\Delta \tilde{s}_t(x_t) & \text{under } R_\sigma \\ \gamma\|\nabla \tilde{s}_t(x_t)\|^2 - \gamma\big(\nabla \tilde{s}_t(x_t) \cdot x_t\big) + \partial_t \tilde{s}_t(x_t) - \gamma \Delta \tilde{s}_t(x_t) & \text{under } R_\gamma \end{cases}$$

Step 2 (Simulate Forward Coupling $\pi^{F,(k)}$): Drawing $x_1 \sim p_1$, we integrate the learned backward SDE from $t = 1$ down to $t = 0$ (with positive reverse step $\mathrm{d}\tau = -\mathrm{d}t$):

$$\mathrm{d}X_t = \begin{cases} -\sigma^2 \nabla \tilde{s}_t^{B,(k)}(X_t)\,\mathrm{d}\tau + \sigma\,\mathrm{d}\bar{W}^P_\tau & \text{under } R_\sigma \\ -\big(-\gamma X_t + 2\gamma \nabla \tilde{s}_t^{B,(k)}(X_t)\big)\mathrm{d}\tau + \sqrt{2\gamma}\,\mathrm{d}\bar{W}^P_\tau & \text{under } R_\gamma \end{cases}$$

Step 3 (Train Forward Network $s^{F,(k)}$): Minimizing L^F_{SB} over the newly generated forward coupling $\pi^{F,(k)}$ takes the symmetric form:

$$s^{F,(k)} = \mathrm{argmin}_{s} \left\{ \mathbb{E}_{t,\pi^{F,(k)},q_{t|01}} \left[s_0(x_0) - s_1(x_1) + \mathcal{I}^F(s_t, x_t) \right] \right\}$$

where the forward integrand $\mathcal{I}^F$ differs solely in the positive sign of the Laplacian term:

$$\mathcal{I}^F(s_t, x_t) = \begin{cases} \dfrac{\sigma^2}{2}\|\nabla s_t(x_t)\|^2 + \partial_t s_t(x_t) + \dfrac{\sigma^2}{2}\Delta s_t(x_t) & \text{under } R_\sigma \\ \gamma\|\nabla s_t(x_t)\|^2 - \gamma\big(\nabla s_t(x_t) \cdot x_t\big) + \partial_t s_t(x_t) + \gamma \Delta s_t(x_t) & \text{under } R_\gamma \end{cases}$$

Step 4 (Simulate Backward Coupling $\pi^{B,(k+1)}$): Drawing $x_0 \sim p_0$, we integrate the learned forward SDE from $t = 0$ up to $t = 1$:

$$\mathrm{d}X_t = \begin{cases} \sigma^2 \nabla s_t^{F,(k)}(X_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W^P_t & \text{under } R_\sigma \\ \big(-\gamma X_t + 2\gamma \nabla s_t^{F,(k)}(X_t)\big)\mathrm{d}t + \sqrt{2\gamma}\,\mathrm{d}W^P_t & \text{under } R_\gamma \end{cases}$$

Inference and Sampling

Upon convergence after K outer iterations, the algorithm yields the optimized forward and backward action networks, denoted for brevity as $s_t^F := s_t^{F,(K-1)}$ and $\tilde{s}_t^B := \tilde{s}_t^{B,(K-1)}$. Because both scalar potentials are available, the trained model supports bidirectional stochastic transport, deterministic ODE sampling, and direct marginal density evaluation.

To generate stochastic trajectories transporting samples from the source distribution p_0 to the target p_1, we draw $x_0 \sim p_0$ and integrate the learned forward SDE from $t = 0$ to $t = 1$:

$$\mathrm{d}X_t = \big(f_t(X_t) + \sigma_t^2 \nabla s_t^F(X_t)\big)\mathrm{d}t + \sigma_t \mathrm{d}W_t, \qquad X_0 = x_0$$

Conversely, to sample in the reverse direction from p_1 to p_0, we draw $x_1 \sim p_1$ and integrate the learned backward SDE over the positive reverse clock $\tau = 1 - t \in [0, 1]$ (corresponding to physical time t running from 1 down to 0):

$$\mathrm{d}X_t = -\big(f_t(X_t) + \sigma_t^2 \nabla \tilde{s}_t^B(X_t)\big)\mathrm{d}\tau + \sigma_t \mathrm{d}\bar{W}_\tau, \qquad X_1 = x_1$$

In practice, both SDEs are discretized over M uniform time steps using the Euler–Maruyama scheme.

At the fixed point, the optimal forward drift $u_t^F(x) = f_t(x) + \sigma_t^2 \nabla s_t^F(x)$ and backward drift $u_t^B(x) = f_t(x) + \sigma_t^2 \nabla \tilde{s}_t^B(x)$ satisfy Nelson's kinematic relation:

$$u_t^F(x) - u_t^B(x) = \sigma_t^2 \nabla \log p_t(x) \implies \nabla \log p_t(x) = \nabla s_t^F(x) - \nabla \tilde{s}_t^B(x)$$

Substituting this identity into the forward Fokker–Planck equation $\partial_t p_t = -\nabla \cdot (p_t u_t^F) + \frac{\sigma_t^2}{2} \Delta p_t$ eliminates the second-order diffusion operator, yielding the deterministic Probability Flow ODE:

$$\frac{\mathrm{d}X_t}{\mathrm{d}t} = \frac{1}{2}\big(u_t^F(X_t) + u_t^B(X_t)\big) = f_t(X_t) + \frac{\sigma_t^2}{2}\Big(\nabla s_t^F(X_t) + \nabla \tilde{s}_t^B(X_t)\Big)$$

Specializing this ODE to our two reference measures gives:

$$\frac{\mathrm{d}X_t}{\mathrm{d}t} = \begin{cases} \dfrac{\sigma^2}{2}\Big(\nabla s_t^F(X_t) + \nabla \tilde{s}_t^B(X_t)\Big) & \text{under } R_\sigma \\ -\gamma X_t + \gamma\Big(\nabla s_t^F(X_t) + \nabla \tilde{s}_t^B(X_t)\Big) & \text{under } R_\gamma \end{cases}$$

This ODE induces the exact same time-marginal densities $(p_t)_{t\in[0,1]}$ as the stochastic bridge while enabling accelerated sampling via adaptive numerical ODE solvers, deterministic bidirectional coupling ($X_0 \leftrightarrow X_1$), and exact likelihood computation via the instantaneous change-of-variables formula.

Algorithm 2: Iterative Action Matching for Schrödinger Bridges

Input: Boundary datasets p_0 and p_1, analytical reference bridge $q_{t|01}(x_t \mid x_0, x_1)$, reference drift $f_t(x)$ and volatility σ_t, number of iterations $K \in \mathbb{N}$

Initialize the backward empirical coupling by drawing independent samples (or via mini-batch optimal transport): $\pi^{B,(0)}(x_0, x_1) = p_0(x_0)p_1(x_1)$.

for $k = 0$ **to** $K - 1$ **do**

Step 1 (Train Backward): Update network $\tilde{s}^{B,(k)}$ by minimizing the backward objective:

$$\tilde{s}^{B,(k)} = \operatorname{argmin}_{\tilde{s}} \left\{ \mathbb{E}_{t,\pi^{B,(k)},q_{t|01}} \left[\tilde{s}_0(x_0) - \tilde{s}_1(x_1) + \frac{\sigma_t^2}{2} \|\nabla \tilde{s}_t\|^2 + \nabla \tilde{s}_t \cdot f_t + \partial_t \tilde{s}_t - \frac{\sigma_t^2}{2} \Delta \tilde{s}_t \right] \right\}$$

Step 2 (Simulate Forward): Draw $x_1 \sim p_1$ and integrate the backward SDE from $t = 1$ down to 0:

$$\mathrm{d}X_t = -\big(f_t(X_t) + \sigma_t^2 \nabla \tilde{s}_t^{B,(k)}(X_t)\big)\mathrm{d}\tau + \sigma_t \mathrm{d}\bar{W}_\tau^P, \qquad X_1 = x_1$$

Store the resulting simulated initial states as pairs (X_0, x_1) to form the forward coupling $\pi^{F,(k)}$.

Step 3 (Train Forward): Update network $s^{F,(k)}$ by minimizing the forward objective:

$$s^{F,(k)} = \operatorname{argmin}_{s} \left\{ \mathbb{E}_{t,\pi^{F,(k)},q_{t|01}} \left[s_0(x_0) - s_1(x_1) + \frac{\sigma_t^2}{2} \|\nabla s_t\|^2 + \nabla s_t \cdot f_t + \partial_t s_t + \frac{\sigma_t^2}{2} \Delta s_t \right] \right\}$$

Step 4 (Simulate Backward): Draw $x_0 \sim p_0$ and integrate the forward SDE from $t = 0$ up to 1:

$$\mathrm{d}X_t = \big(f_t(X_t) + \sigma_t^2 \nabla s_t^{F,(k)}(X_t)\big)\mathrm{d}t + \sigma_t \mathrm{d}W_t^P, \qquad X_0 = x_0$$

Store the resulting simulated terminal states as pairs (x_0, X_1) to form the backward coupling $\pi^{B,(k+1)}$.

Output: Optimized forward and backward networks $s^{F,(K-1)}$ and $\tilde{s}^{B,(K-1)}$